FIT @ HCMUS · Department of Software Engineering

CS423 / CSC15003 — SOFTWARE TESTING

AI-augmented · 2026 Edition

Course Syllabus

Lecturer: Dr. Lam Quang Vu

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About This Course

Position in curriculum

- Major Selective course of the Software Engineering specialisation (KTPM).
- 4 credits = 45 lecture hours + 30 lab hours, taught across 11 weeks.
- Prerequisites: OOP, Databases, Intro to Software Engineering, Web Development.
- Suitable for students aiming at QA / Test Engineer / DevOps / SDET roles.
- **2026 edition: AI-augmented — every learning outcome includes AI collaboration.**

References

- ISTQB Foundation Level Syllabus (latest).
- Pressman & Maxim (2014) — Software Engineering: A Practitioner's Approach.
- Myers et al. (2011) — The Art of Software Testing.
- Hardman (2025) — Post-AI Bloom Taxonomy.
- Kharbach (2026) — AI Use Policy Templates for Higher Education.

Course Objectives — 2026 AI-first Revision

Traditional Outcomes (G1–G8)

G1. Concepts + STLC + V-model.
G2. Black-box design (domain, decision table, state, use case).
G3. White-box design (stmt, branch, MC/DC).
G4. Test plan / case / bug report per IEEE 829.
G5. UI automation (Selenium / Playwright).
G6. Performance testing (Load/Stress/Spike).
G7. API testing (Postman + Newman).
G8. GUI + Usability + a11y testing.

NEW: AI-Applied Outcomes (G9.1–G9.6)

G9.1 UNDERSTAND — critique AI testing explanations.
G9.2 APPLY — collaborate with AI to draft test cases.
G9.3 ANALYSE — audit AI artifacts (V/I/IC).
G9.4 COLLABORATE — pair AI + human in a team workflow.
G9.5 CREATE — design AI tools / plugins / harnesses.
G9.6 DISRUPT — propose Continuous-AI testing workflows.

Ref: Post-AI Bloom Taxonomy (Hardman 2025).

Every HW + Project carries BOTH traditional and AI CLOs. Exams test both.

Grading System

HW (1–6) **30%**	Individual · Every HW has AI Critique 200–300 words
Seminar (SM) **20%**	Group · AI for research · Hand-on workshop
Midterm **10%**	Closed-book on paper · NO AI
Final **40%**	Closed-book cumulative · Quizzes

Apply AI in This Course — What's New

AI is part of every learning outcome

Students may use ANY AI tool; the Faculty does NOT provide paid accounts.

- 6 Homeworks (HW01–HW06) released across the term; every HW is AI-augmented.
- Use ANY AI Tool — e.g., ChatGPT, Claude, Gemini, Copilot, Cursor. Declare what you used.
- 6 new Bloom-AI CLOs (G9.1–G9.6) sit alongside the traditional CLOs (G1–G8).
- **Every AI-assisted submission attaches: [AI-02] Audit Report + [AI-03] Disclosure + [AI-05] Checklist.**
- Closed-book exams (Midterm + Final) — NO AI of any kind.
- Random 10–30% of students invited to a 5–7-min oral defense the week after each HW.

AI Use Categories — 5 tiers (which applies?)

Cat. 1 – No AI	Midterm · Final · Week-4 Domain Testing paper workshop
Cat. 2 – AI for Preparation	Brainstorm, study questions — NOT generating final work
Cat. 3 – AI Feedback/Revision	Critique / polish AFTER you have a complete draft
Cat. 4 – AI-Assisted Production	AI generates parts; YOU audit + integrate (most HWs)
Cat. 5 – AI-Integrated Work	AI as core methodology (Project · Seminar · Capstone)

Bloom-AI Application Levels (1/2) — G9.1, G9.2, G9.3

G9.1 UNDERSTAND	Ask AI to explain a concept; identify ≥ 3 errors using the spec/ISTQB.	**Applied in:** HW01 · all HW
G9.2 APPLY	Use AI to draft test cases / scripts / code. Then run + refine yourself.	**Applied in:** HW02 · HW04 · HW06
G9.3 ANALYSE	Audit AI artifacts: V/I/IC. Find missed boundaries.	**Applied in:** HW02 · HW04 · HW05 · HW06

Bloom-AI Application Levels (2/2) — G9.4, G9.5, G9.6

G9.4 **COLLABORATE**	Pair AI + human in a team workflow. Document who did what.	**Applied in:** HW04 · HW05 · HW06
G9.5 **CREATE**	Design a NEW AI tool / plugin / evaluation harness for testing.	**Applied in:** HW06
G9.6 **DISRUPT**	Propose a Continuous-AI workflow that changes the testing process.	**Applied in:** HW05 · Final

AI Templates — required submissions (/AI Templates folder)

[AI-01]	**AI Agreement**	Read at start of term — defines 5 categories + policy
[AI-02]	**AI Audit Report**	5-section template per artifact — REQUIRED with every HW
[AI-03]	**AI Disclosure Form**	Per-assignment — sign + attach to every AI-assisted HW
[AI-04]	**Reflective Statement**	Required for Seminar
[AI-05]	**Privacy Checklist**	Self-check before submission · required with every HW
[AI-06]	**Student Acknowledgement**	Sign in Week 1 · required before AI-assisted HW is graded

Both English (_En) and Vietnamese (_Vn) versions in the AI Templates folder.

AI Audit Report – 5 required sections per AI artifact

1	**Prompt + Tool**	Full prompt + tool name (Claude/GPT/Cursor/Diffblue) + timestamp
2	**AI Output**	Verbatim AI output (no paraphrase) + annotated screenshot
3	**Verdict**	VALID / INVALID / INCOMPLETE – with reasoning grounded in ISTQB
4	**Reasoning**	2–5 sentences referencing the matching ISTQB slide / section
5	**Student fix**	Revised test case / script / checklist – highlight changes

Mandatory Disclosure – standard template

- [REQUIRED] Every HW + Project must include a disclosure paragraph at the end
- **Template:**
  - — "This test case/script/dataset/report was initially generated by [AI tool];
  - — I reviewed it, modified [section X], and added [edge cases Y, Z];
  - — section [W] was written entirely by myself.
  - — The detailed AI Audit Report is attached as Appendix A."
- Missing disclosure = academic misconduct
- False disclosure (understating AI use to get higher grade) = equivalent to plagiarism

Random 30% oral defenses the week after each deadline verify disclosure honesty.

SUT — EShop (for HW02–HW06 + Project)

Course-wide SUT: EShop — a Vietnamese e-commerce demo built for software-testing education.

Repository: https://github.com/ttbhanh/eshop-sut

Architecture (4 sub-systems)		
Backend API	Node.js + Express + SQLite	*http://localhost:3000*
Frontend Web	React + Vite + Tailwind	*http://localhost:5173*
Web Admin	React + Vite + Tailwind	*http://localhost:5174*
Mobile App	React Native + Expo	*LAN IP of server*

22 Functional Requirements + 7 Security Requirements (full spec in README.md)

- FR-01..04 Authentication (Register, Login + 3-fail lockout, Forgot password OTP, Profile)
- FR-05..09 Products + Cart + Checkout + Coupon (5 conditions)
- FR-10..11 Order State Machine (pending → confirmed → shipping → delivered)
- FR-12..19 Admin (Dashboard, CRUD, CSV Import RFC 4180, User mgmt)
- FR-20..24 Mobile + GUI standards + Form + Navigation + Feedback
- SEC-01..07 Security (no plaintext password, JWT, role check, escape, parameterised SQL)

Default test accounts: admin@eshop.com / Admin123! · test@eshop.com / Test1234!

Homework Schedule — 6 HW across Weeks 1 / 3 / 5 / 7 / 9 / 11

Week 1	HW01	QA/QC Jobs · 20 Defects · Test a Physical Product	*5h · G9.1 · G9.3*
Week 3	HW02	EShop: Domain Testing (Personal Seed)	*10h · G9.2 · G9.3*
Week 5	HW03	EShop: GUI & Usability (Real Users)	*10h · G9.3 · G9.4*
Week 7	HW04	EShop: Automation (Hand vs AI)	*12h · G9.2 · G9.3 · G9.4*
Week 9	HW05	EShop: Performance (AI Misinterpretation Hunt)	*12h · G9.2 · G9.3 · G9.4 · G9.6*
Week 11	HW06	EShop: API Testing (AI-spec-to-tests)	*12h · G9.2 · G9.3 · G9.4 · G9.5*

Every HW = Open AI Mode (Cat. 4). Attach [AI-02] + [AI-03] + [AI-05]. Total = 10% of course grade.

Seminar Topics — 10 Tools-Focused Tracks (Teams of 3–4)

Each team surveys 1 traditional tool + at least 1 AI-augmented tool. Workflow: propose → instructor approves → deep study → user-guide + demo → share with class ≥ 3 days before seminar.

T1 — **GUI & Usability Testing**
Maze · Hotjar · BrowserStack
AI: heuristic review, screenshot diff

T2 — **Web Automation Testing**
Playwright · Selenium · Cypress
AI: Mabl/Testim auto-heal, AI test gen

T3 — **Mobile Automation Testing**
Appium · Espresso · Detox
AI: visual diff (Applitools), AI locators

T4 — **Desktop Automation Testing**
WinAppDriver · Pywinauto · TestComplete
AI: image-based recognition

T5 — **Performance Testing**
JMeter · k6 · Locust
AI: scenario synthesis from logs

T6 — **API & Contract Testing**
Postman · RestAssured · Karate
AI: tests from OpenAPI; Pact AI

T7 — **CI/CD & Test-Harness Engineering**
GitHub Actions · Jenkins · GitLab CI
AI: flaky-test triage, log summarisation

T8 — **Database Testing**
DbUnit · Liquibase · tSQLt
AI: data masking, schema-drift detection

T9 — **Security Testing**
OWASP ZAP · Burp Suite · Semgrep
AI: vulnerability triage, AI fuzzers

T10 — **Mutation Testing & Test Effectiveness**
PIT · Stryker · mutmut
AI: equivalent-mutant detection, AI assertion gen

Anti-Cheat Mechanisms + Oral Defense

Cannot be AI-generated. TAs verify when grading.

- ✗ Photos of physical devices + your student-ID card in the SAME frame (HW01).
- ✗ Videos of test executions with your OWN voice narration (no TTS).
- ✗ Live debug during oral defense: TA changes 1 selector → fix in 10 minutes.
- ✗ Mantis bug reports filed with reporter = your StudentID.
- ✗ Hostname / username watermarks in screenshots ({StudentID}-SWT).
- ✗ EShop test data: <StudentID>_test_<n>@hcmus.edu.vn (verifiable in SQLite DB).
- ✗ Prompt log .md with rolling timestamps — not all dumped at once.

Random 10–30% of students invited to a 5–7-min oral defense the week after each HW.

Assignment Policies — Part 1 (Principles + Format)

General Principles

- All assignments are INDIVIDUAL. Discussion + AI use is encouraged — responsibly.
- Core criteria: smart AI use + methodical testing process.
- Test cases + bugs discovered = basis for grade differentiation.
- Mandatory AI Disclosure per the AI Usage Guideline.

File Format

- Submissions must be TEXT-BASED. Markdown encouraged (LaTeX OK).
- Old Word users → switch to Markdown. Excel summary → into Markdown.
- ALWAYS submit a Save-As-PDF alongside the Markdown source.
- **Submission filename: StudentID_ExerciseID_SelfAssessedGrade.zip**

Version Control

- Use Git (GitHub / Bitbucket / GitLab). Commit per step with clear messages.
- HW01 → personal repo. From HW02 onwards → team repo provided by instructors.

Assignment Policies — Part 2 (Submission + Integrity)

Peer Review + Self-Assessment

- Project work: each student OWNS a SPECIFIC requirement (no overlap with teammates).
- Balance task complexity — too-easy tasks limit the max grade.
- Peer reviews are MANDATORY — find constructive feedback (praise-only = low value).
- Each HW has a self-assessment rubric; self-grade honestly.
- **SelfAssessedGrade in filename = 3 digits, 000–100.**
- Certain exercises carry HIGHER weight than others.

Submission + Deadlines

- Submit via MOODLE.
- **Late submissions: NOT accepted, except critical reasons with documented proof.**
- Max 20 files, each ≤ 20 MB. Use split-and-zip when needed.
- **Misuse / over-reliance on online links in submissions → 0 for the assignment.**

Academic Integrity

- **Any violation of submission rules OR anti-cheat → 0 for the assignment.**

Contact + Tools

Teaching Team

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Course channels

- Moodle — official LMS for submission + announcements
- Moodle Discussion Forum — Q&A (preferred over email)
- FIT Mantis (mantis.fit.hcmus.edu.vn) — bug tracker
- EShop repo — github.com/ttbhanh/eshop-sut (HW02–HW06 SUT)